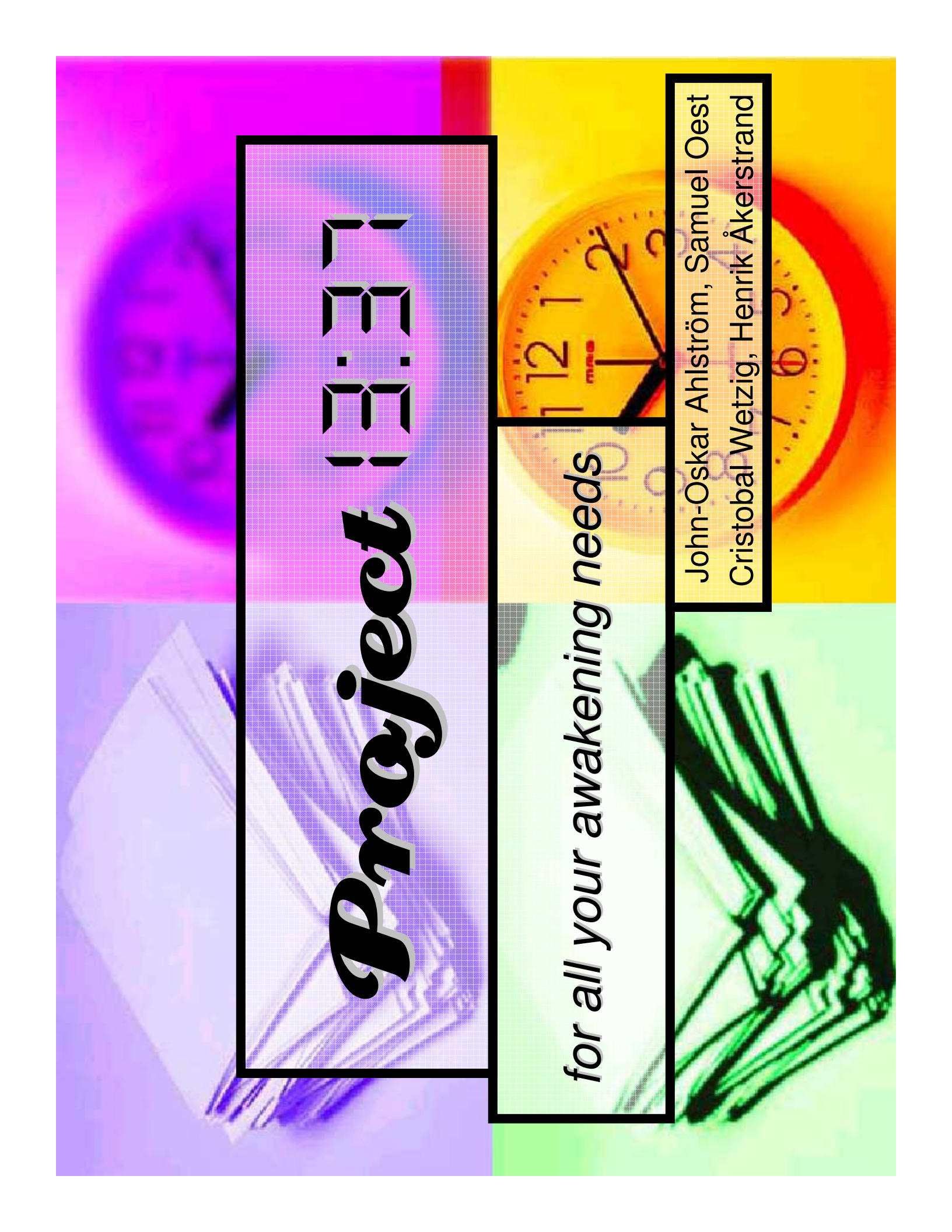
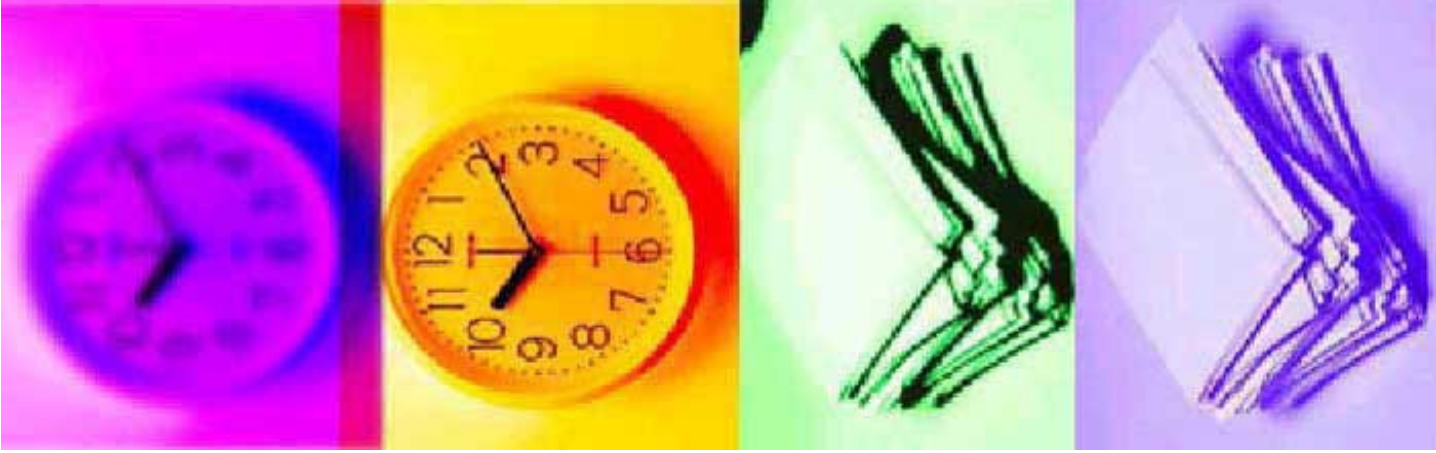




Project 13:33

for all your awakening needs

John-Oskar Ahlström, Samuel Oest
Cristobal Wetzig, Henrik Åkerstrand

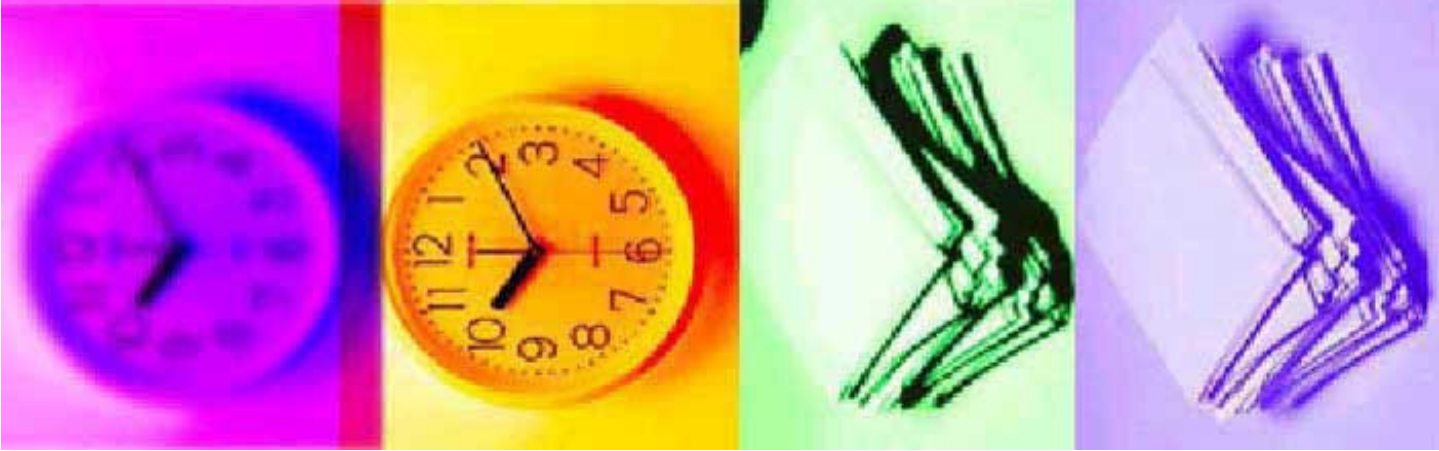


Agenda

- Film!
- Bakgrund och vision
- Komponenter
 - Ljusbmätning
 - Reflexdetektor
 - Ljudet
 - Potentiometrer
 - Styrchip
- Konstruktion
- Tillståndsautomat
- Programmering
- Problem
- Förbättringar
- Saker vi inte hann med
- Frågor

Project 13:37

>>Film

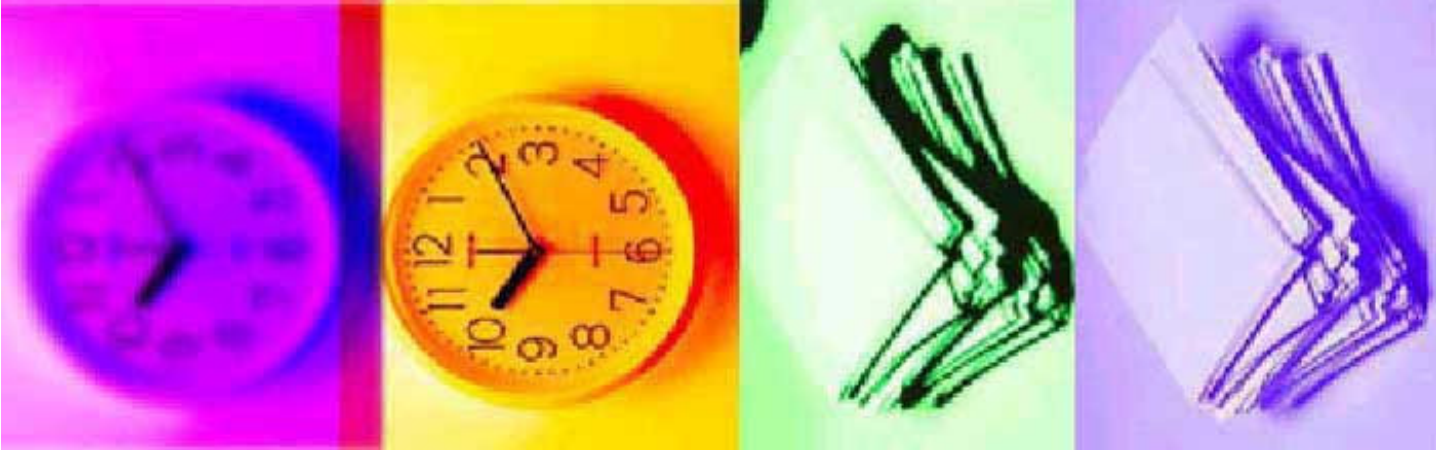


FILM

Agenda<<

Project 13:37

>>Bakgrund



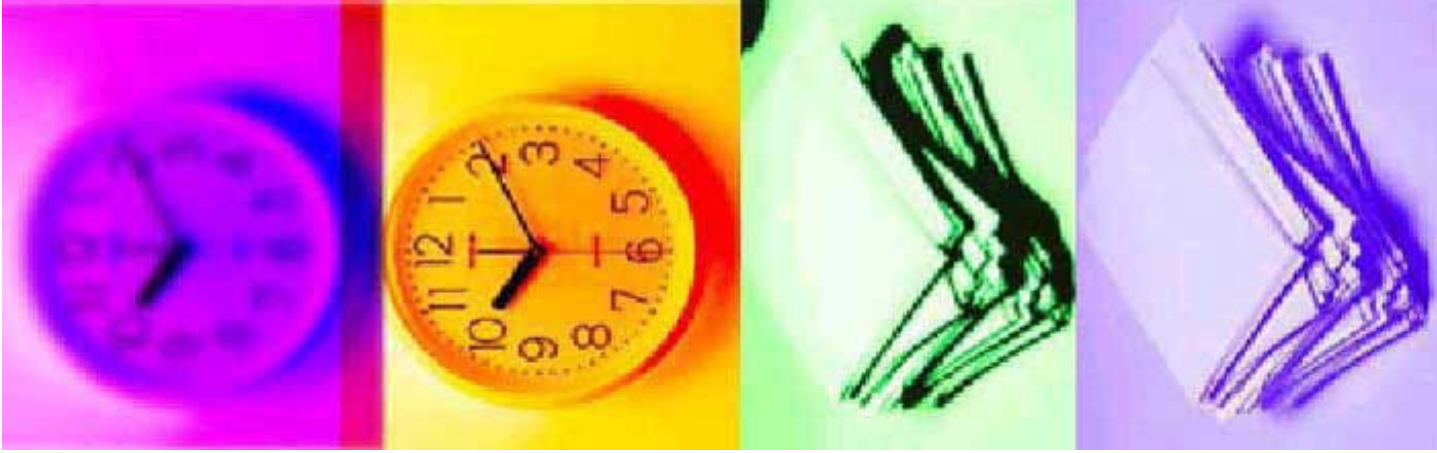
Bakgrund och vision

- Idé
- Kravspecifikation
 - Autonom körning
 - Hinder evasion
 - Väckarklocka
 - Kunna gömma sig
 - Generera ljud
- Angripa utmaningen

Film<<

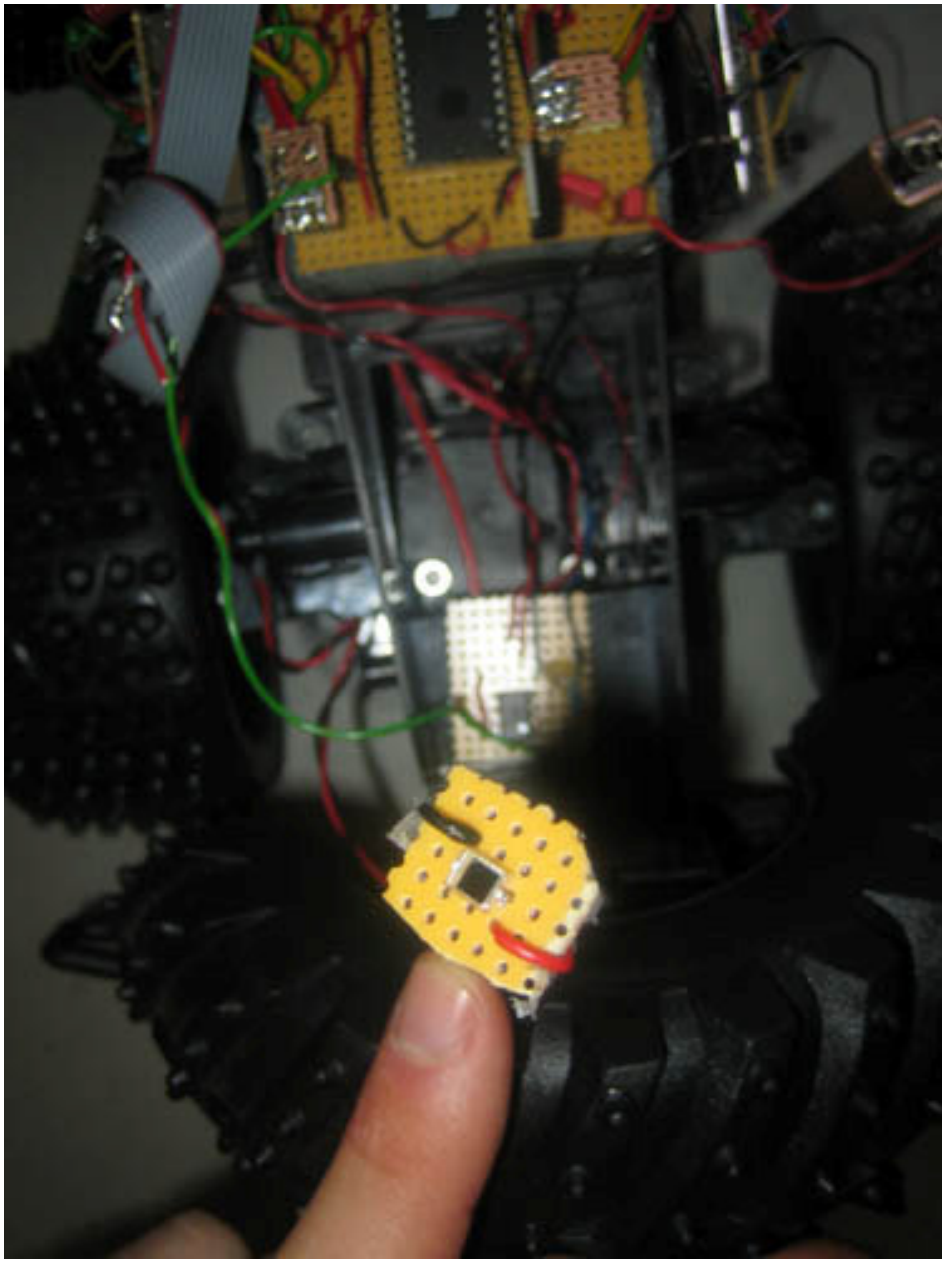
Project 13:37

>>Ljussmätning



Komponenter

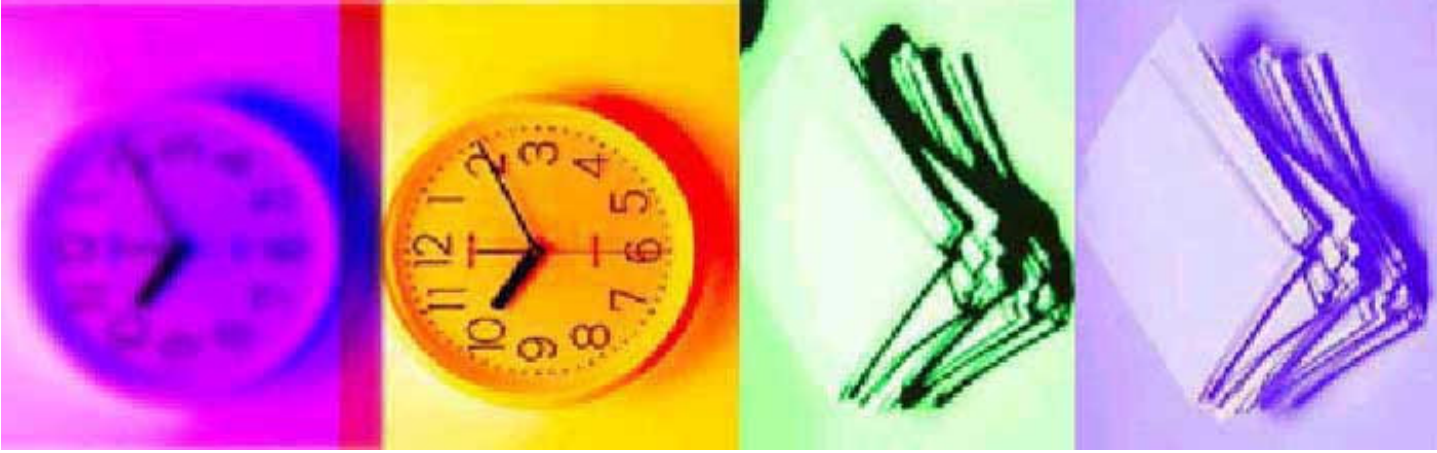
- ljusmätning (BPW 34)



Bakgrund<<

Project 13:37

>>Reflexdetektor



Komponenter

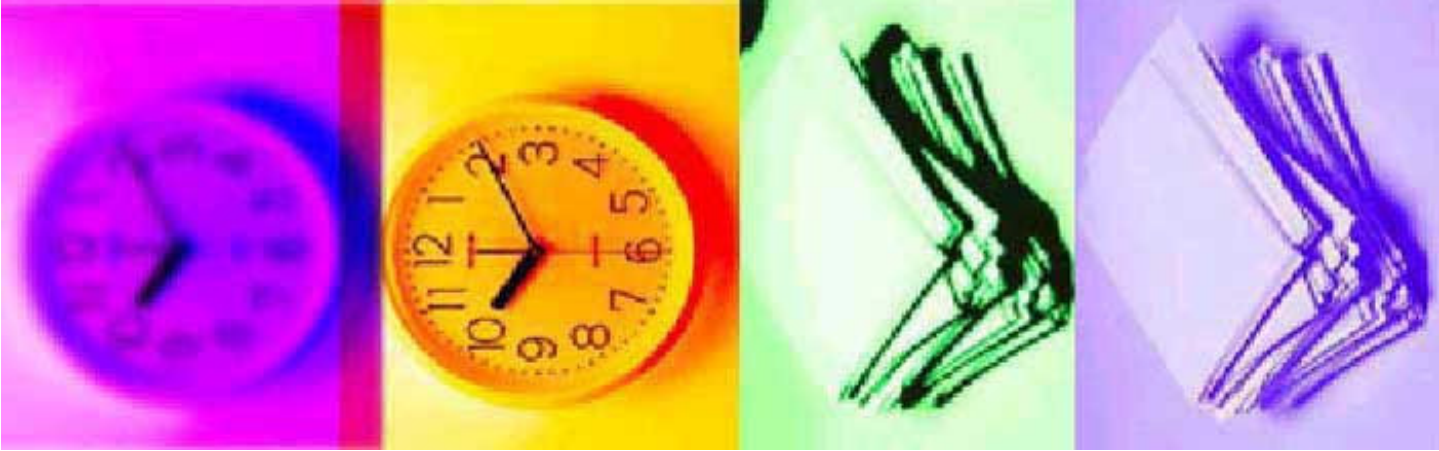
- reflexdetektor (GP 201F)



Ljusmätning<<

Project 13:37

>>Ljudet



Komponenter

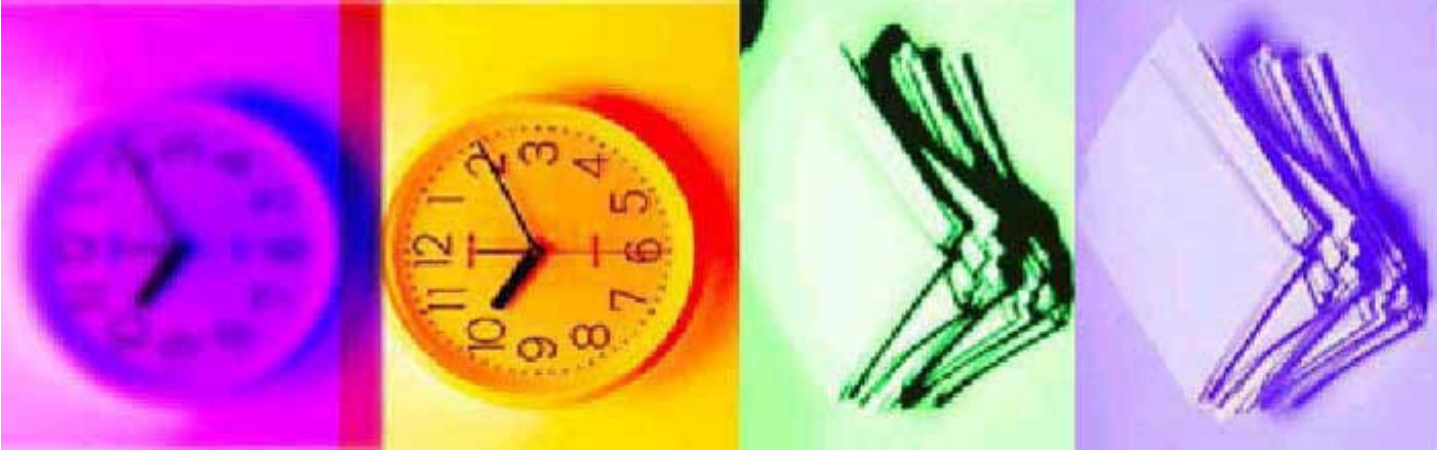
- ljudet



Reflexdetektor<<

Project 13:37

>>Potentiometer



Komponenter

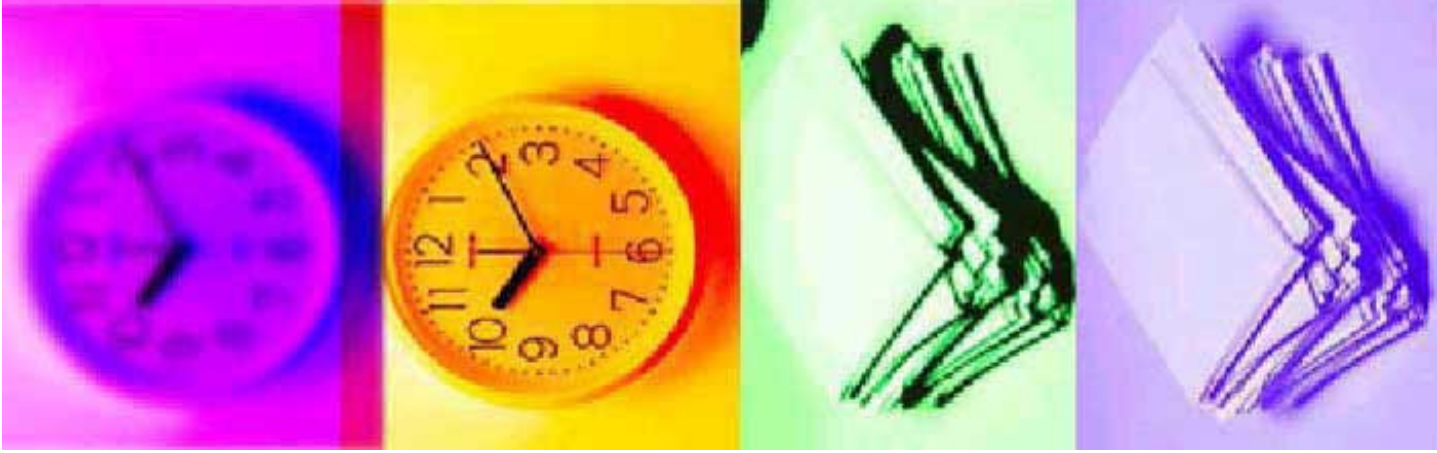
- potentiometer



Ljudet<<

Project 13:37

>>Styrchip



Komponenter

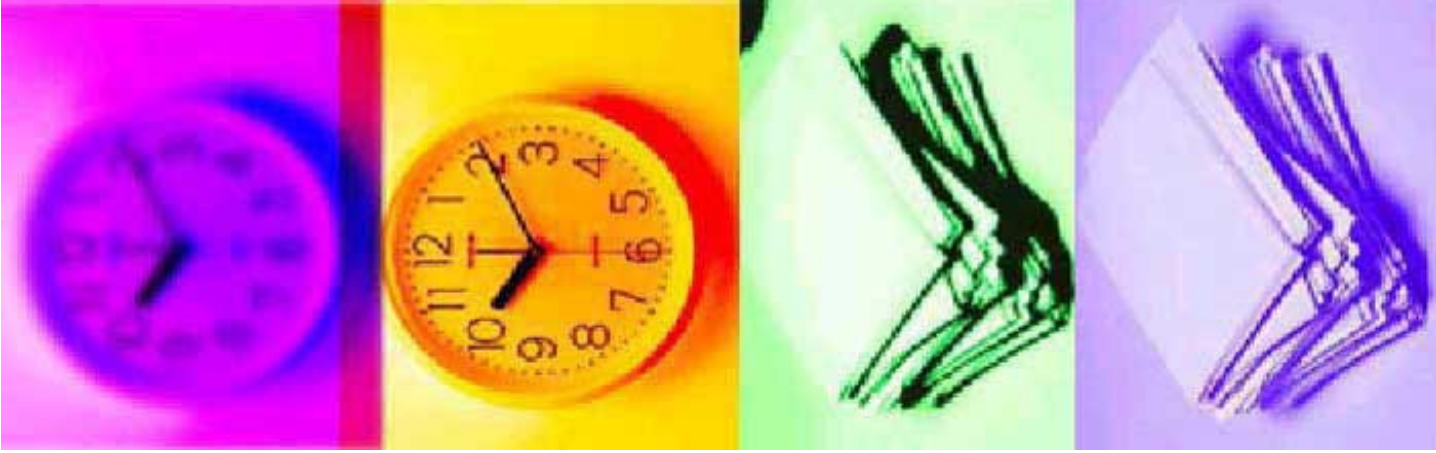
- styrchip (RX2)



Potentiometer<<

Project 13:37

>>Konstruktion



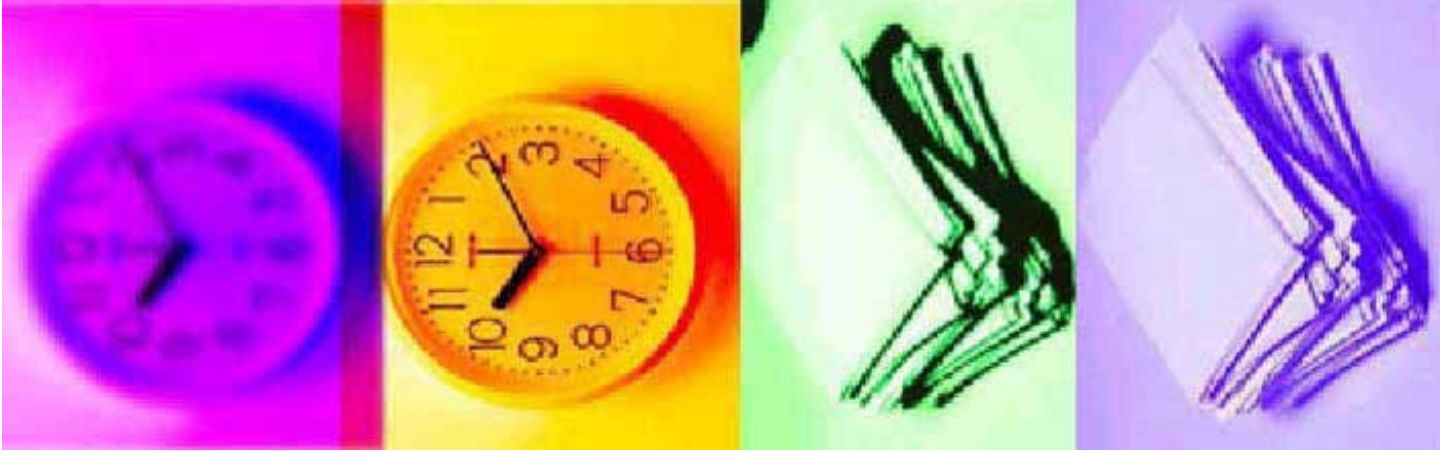
Konstruktion

- Kretsschema
- Modulärt
 - Moderkort
 - Ljusförstärkare
 - Interface till styrkort
 - Reflexdetektor
 - Potentiometrer

Styrchip<<

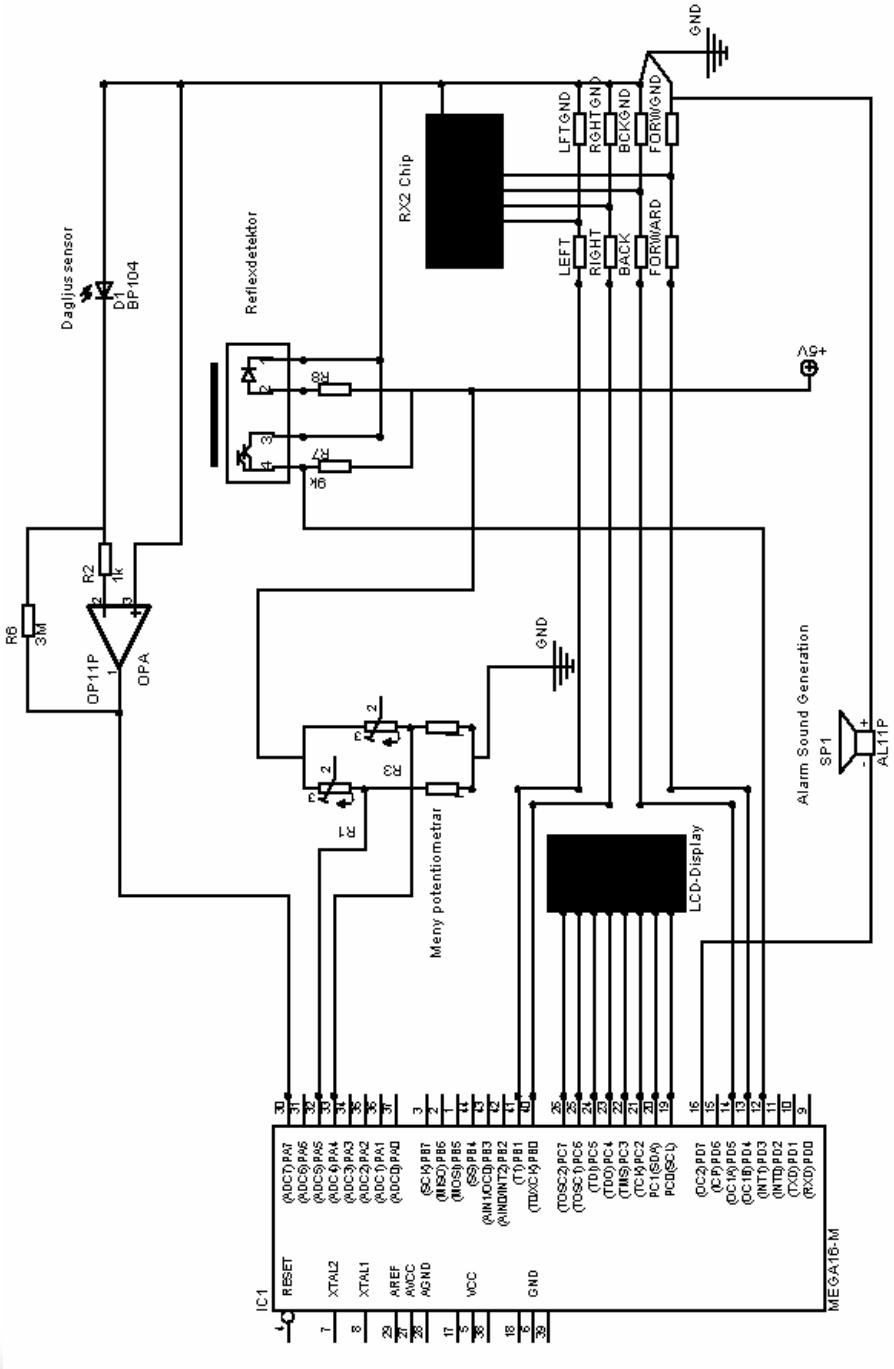
Project 13:37

>>Kretsschema



Konstruktion

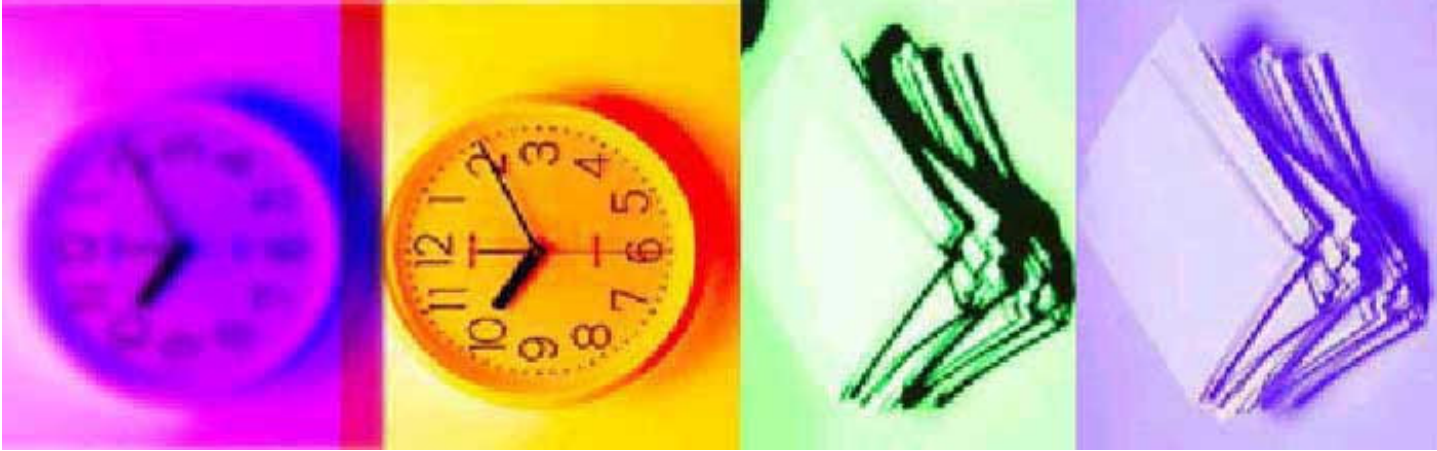
- kretsschema



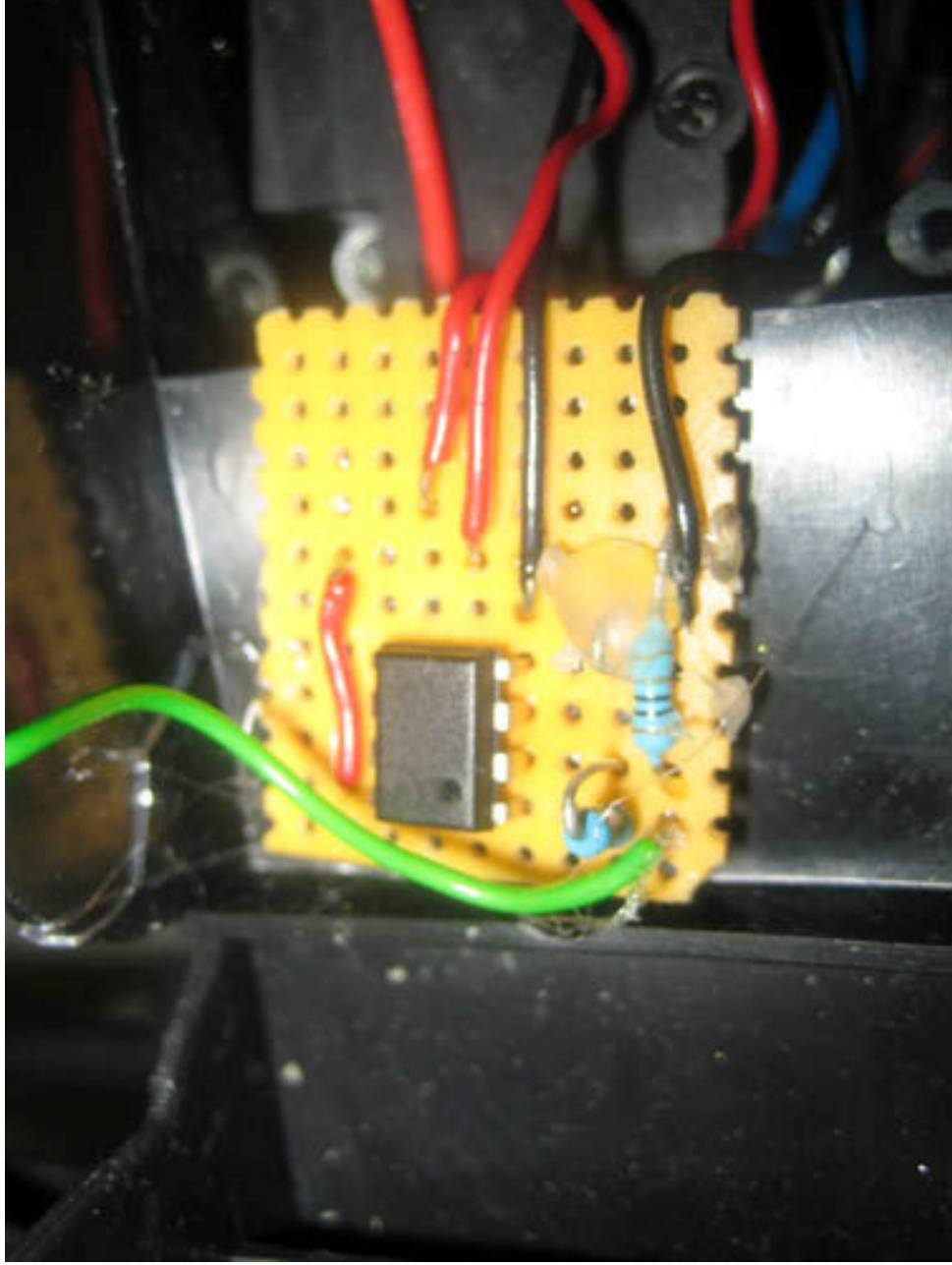
Konstruktion<<

Project 13:37

>>Ljusförstärkare



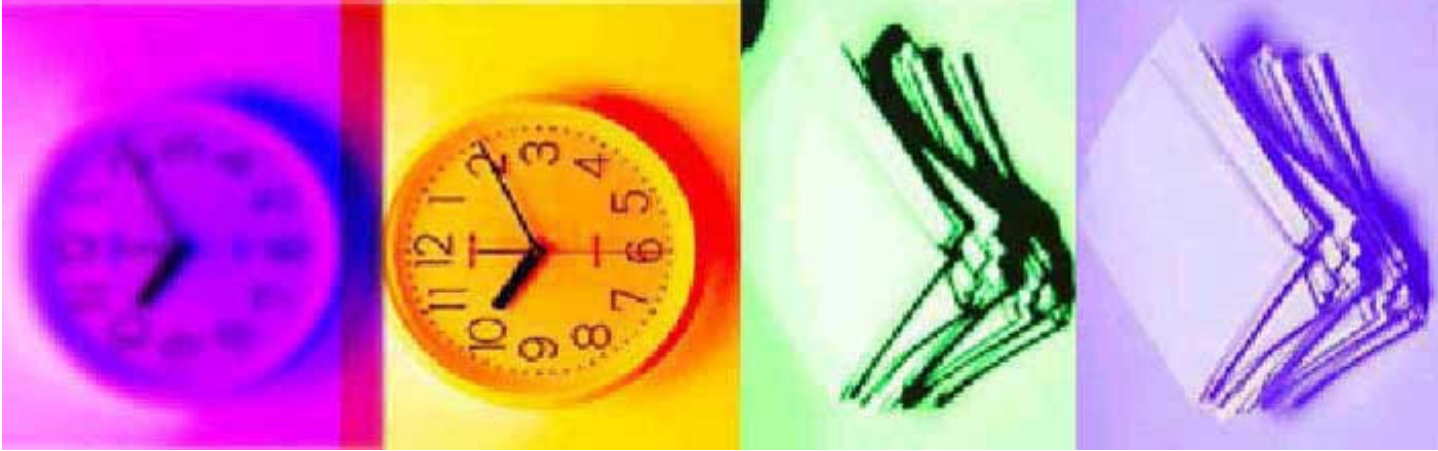
Ljusförstärkare



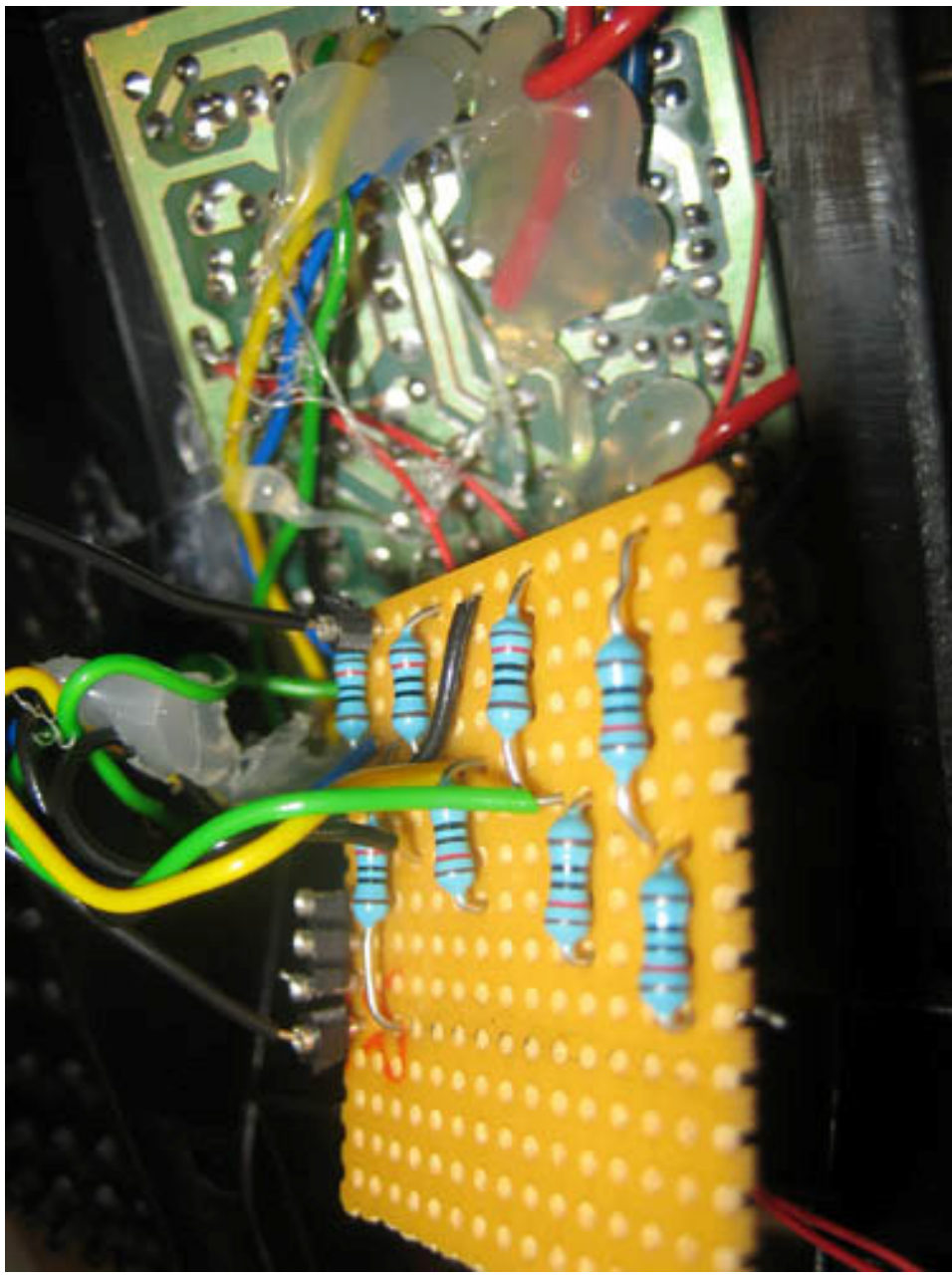
Kretsschema<<

Project 13:37

>>Interface



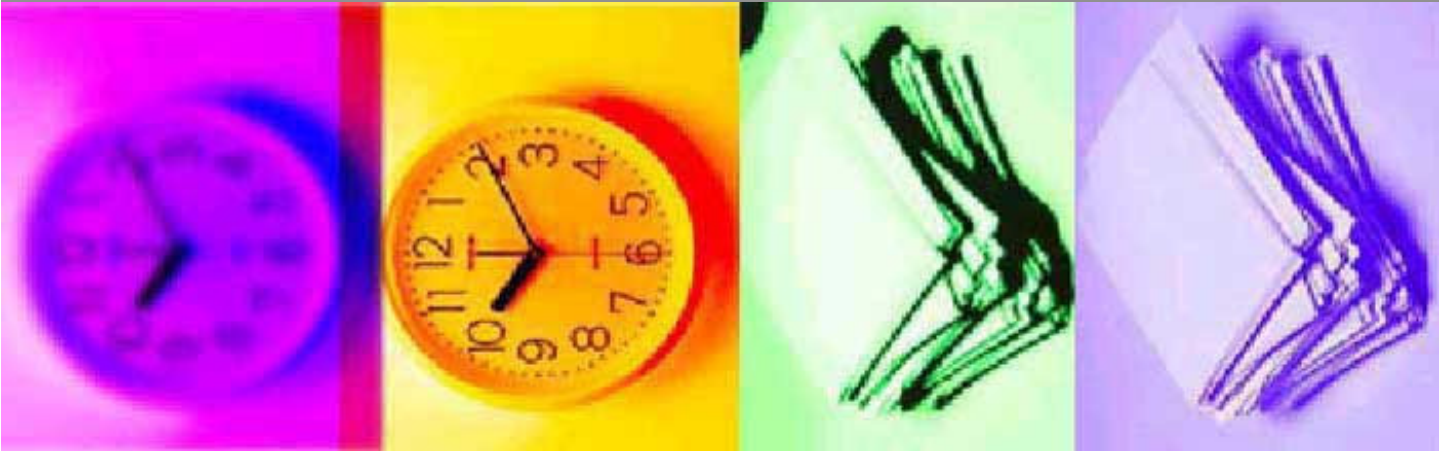
Interface till styrkort



Ljusförstärkare<<

Project 13:37

>>Reflexdetektor



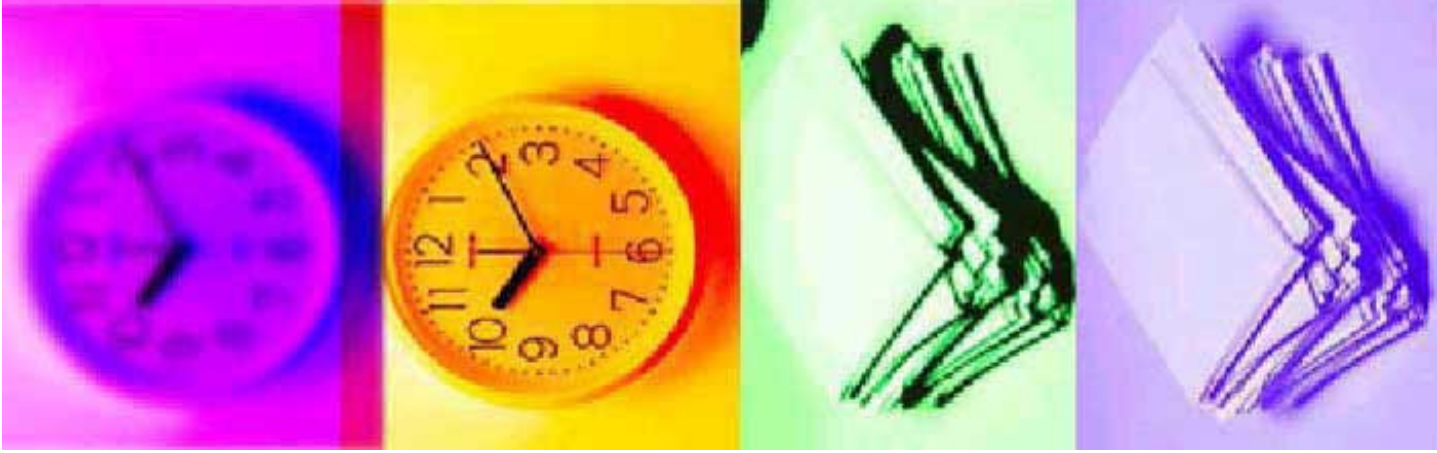
Reflexdetektor



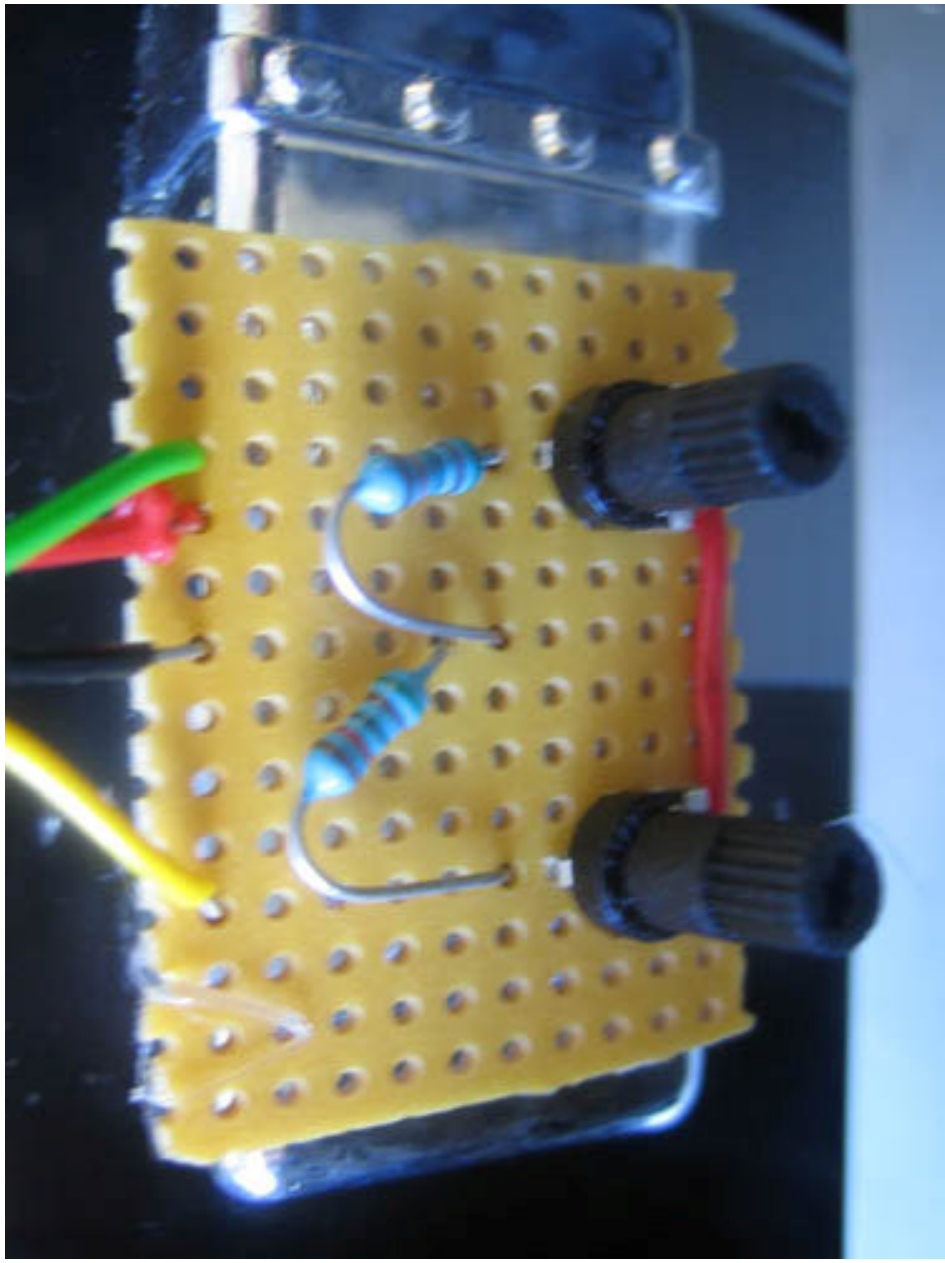
Interface<<

Project 13:37

>>Potentiometer



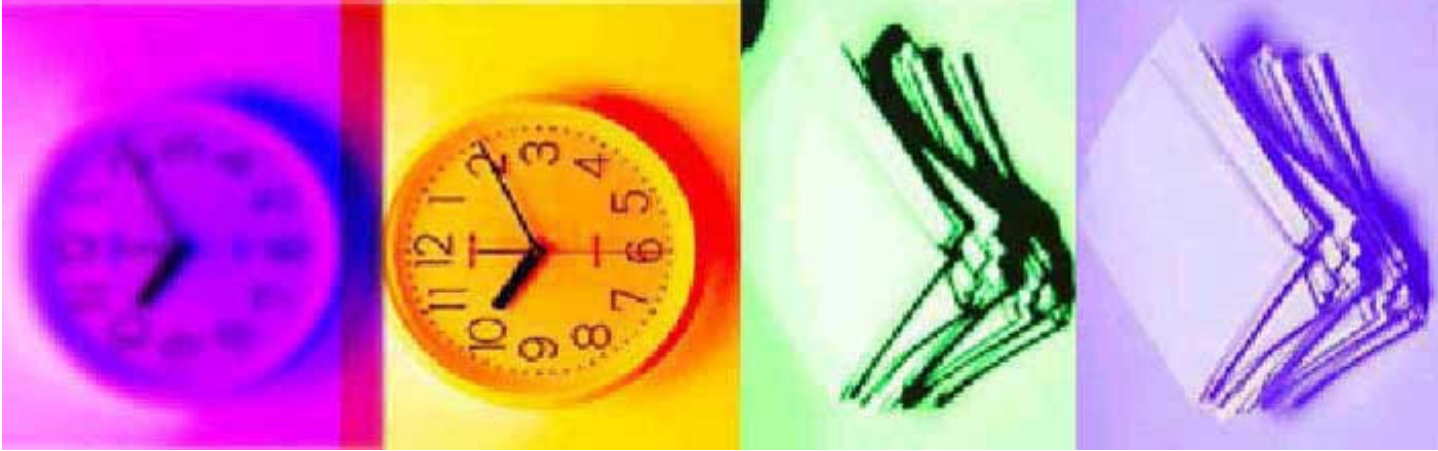
Potentiometer



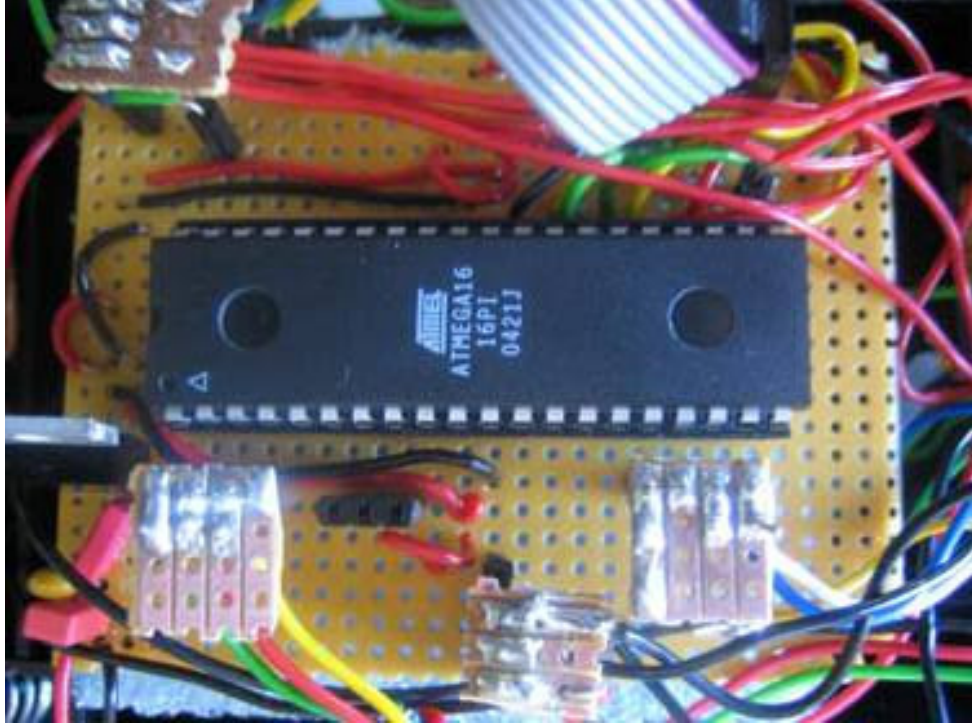
Reflexdetektor<<

Project 13:37

>>Moderkort



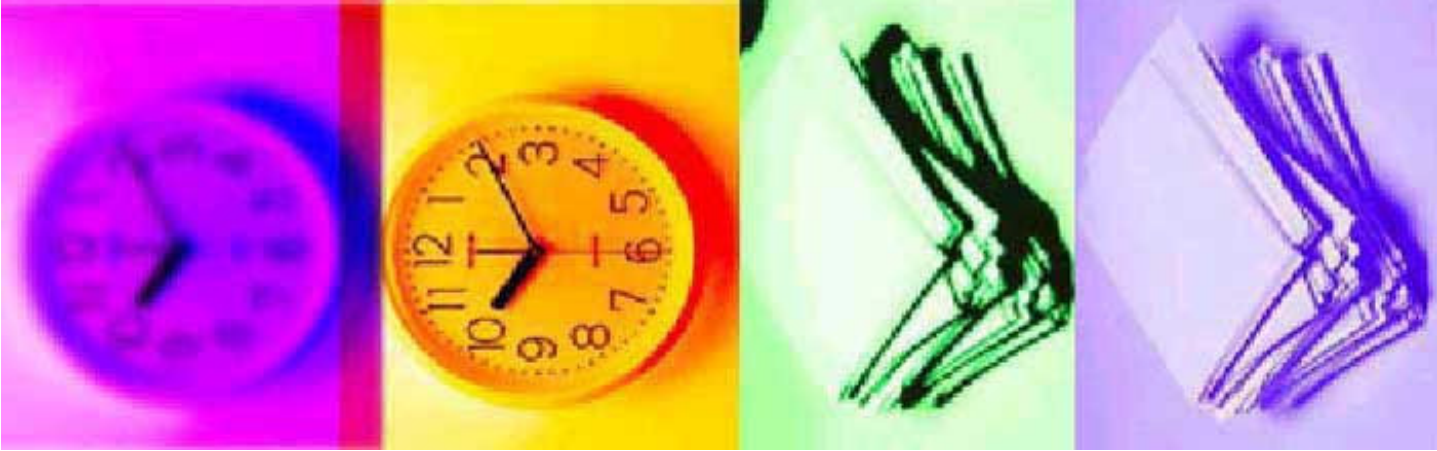
Moderkort



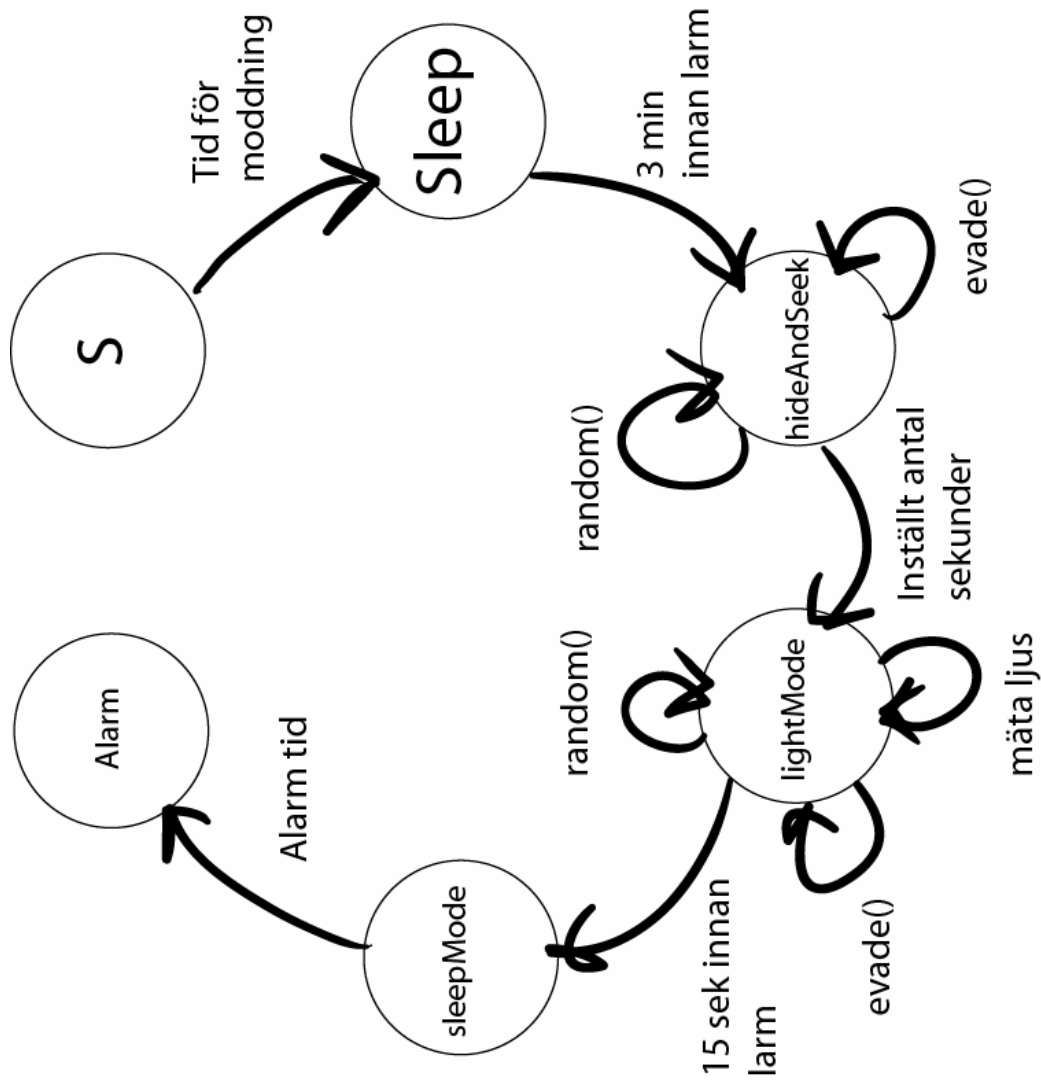
Potentiometer<<

Project 13:37

>>Tillståndsautomat

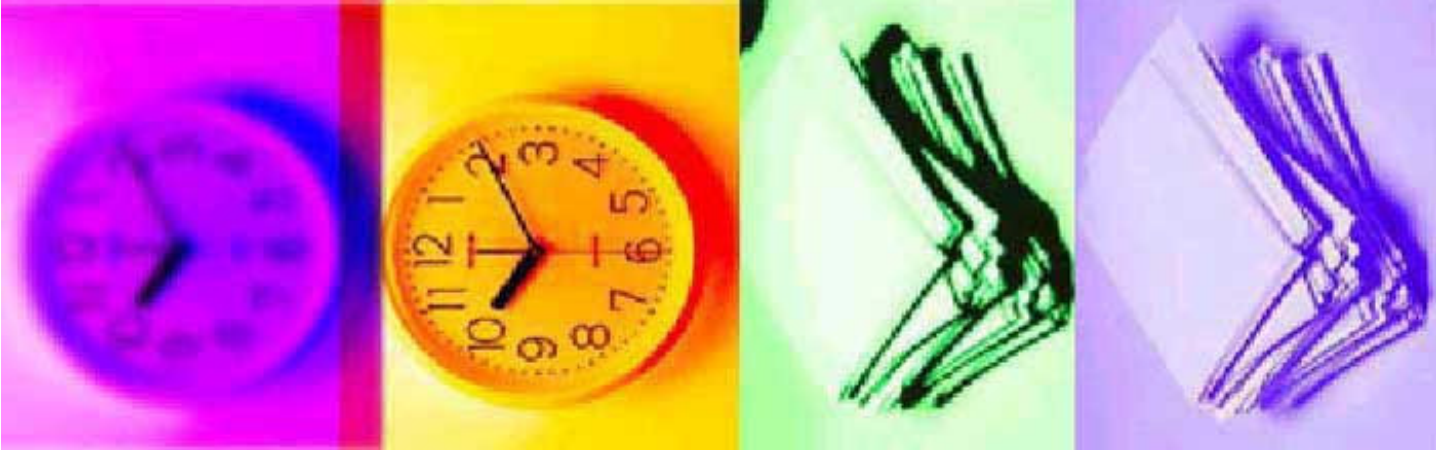


Tillståndautomat



Moderkort<<

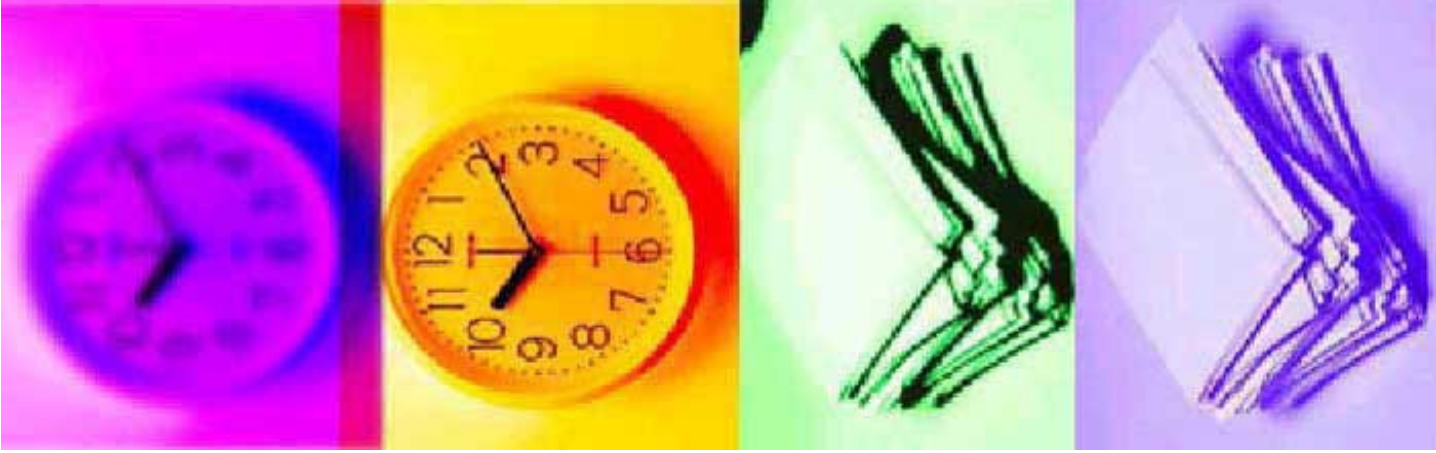
Project 13:37 >> Programmering 1



Programming

- viktiga funktioner

- `startCheck();`
 - `stateToggle();`
 - `hideAndSeek();`
 - `evade();`
 - `random();`
 - `lightMode();`
 - `evade();`
 - `random();`

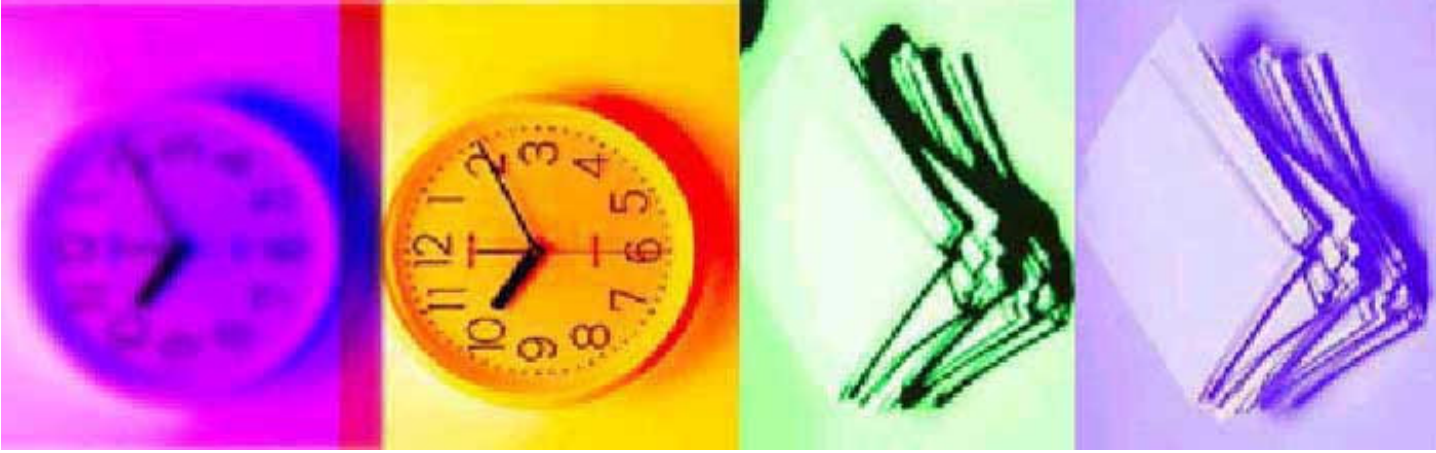


Programming

- interrupts

■ INT1

Reflexdetektor genererar
interrupts



Programmering

- timers

■ TIMER0

- Klocka
- Tidskritiska funktioner och variabler

■ TIMER1

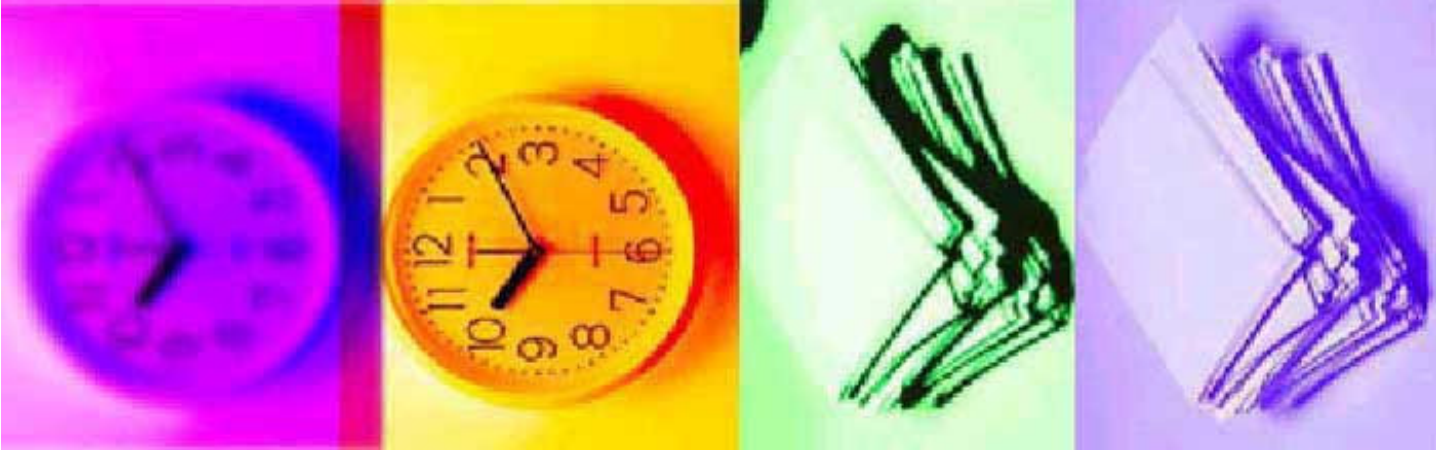
- Pulsbreddsmodulering

■ TIMER2

- Ta in värden från A/D-omvandlare
- Generera ljud

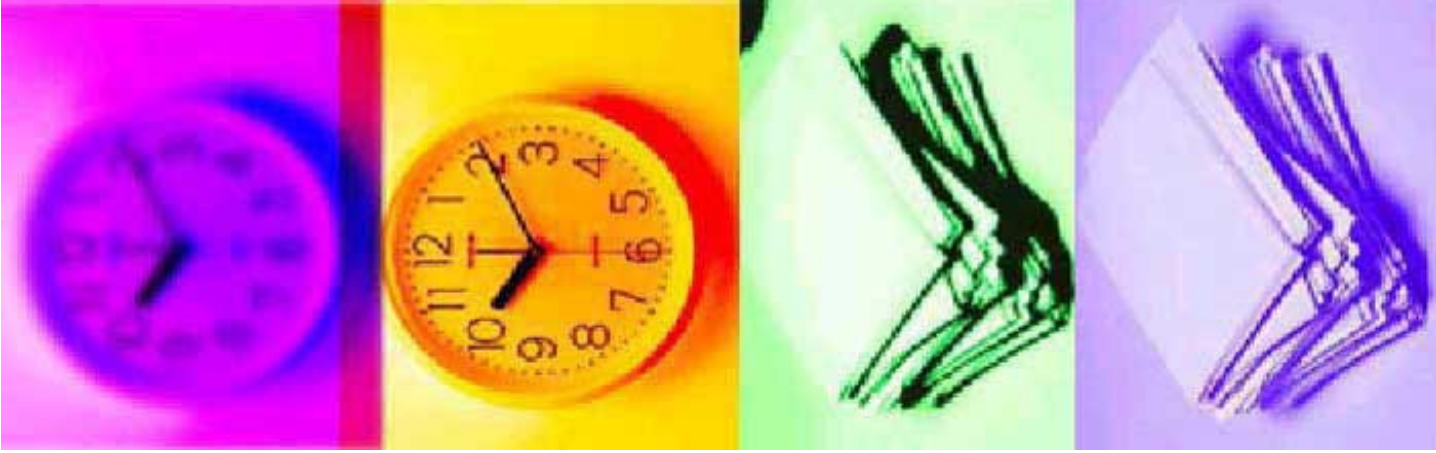
Programmering 2<< **Project** 13:37

>>Problem



Problem

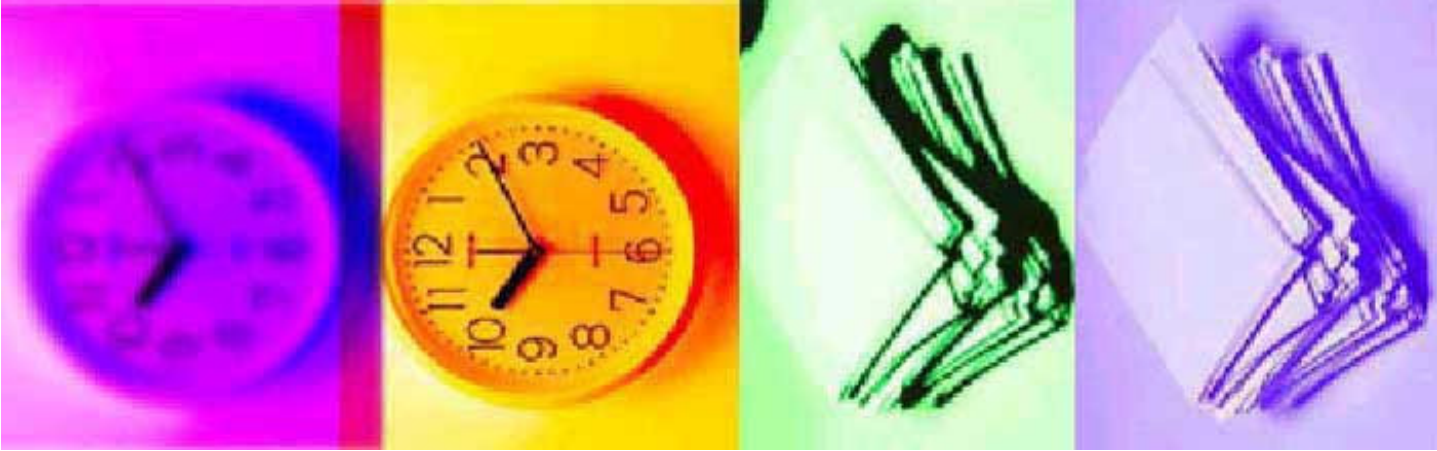
- Gå från utvecklingsmiljö till fungerande prototyp
 - Strömförsörjning
 - Ljusbildning
- Knappar



Förbättringar

- Chassi
- Pre-Evade
- Anpassa hastighet till batteristatus
- Eget fordon (prototyp)
- Spara noder
- Sätta klocka med trådlös anslutning
- Basenhet

Programmering 3<< **Project** 13:37 >>Saker vi inte...



Saker vi inte hann med

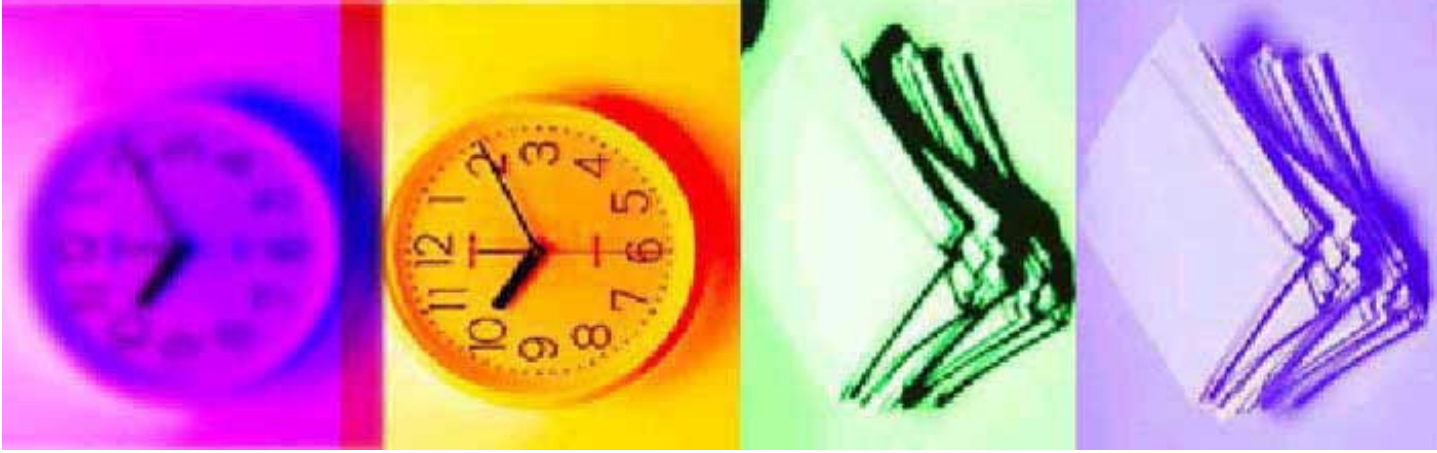
- enligt specifikation

Förbättringar<<

Project 13:37

>>Frågor?

Frågor?



Saker vi inte...<<

Project 13:37